

Ride Cancellation Prediction

CSCI611 Group 5

**Govardhan Reddy B
Tabrez**

Problem Statement

- Uber and Lyft lose millions daily due to unexpected ride cancellations
- NYC TLC publishes detailed taxi trip records publicly every month
- These records include pickup time, location, distance
- But cancellations are never recorded when a ride cancels, there is no trace
- Drivers lose income when rides cancel unexpectedly
- Passengers are left waiting with no warning
- Our goal - build a model that flags a trip as high-risk before the driver is sent

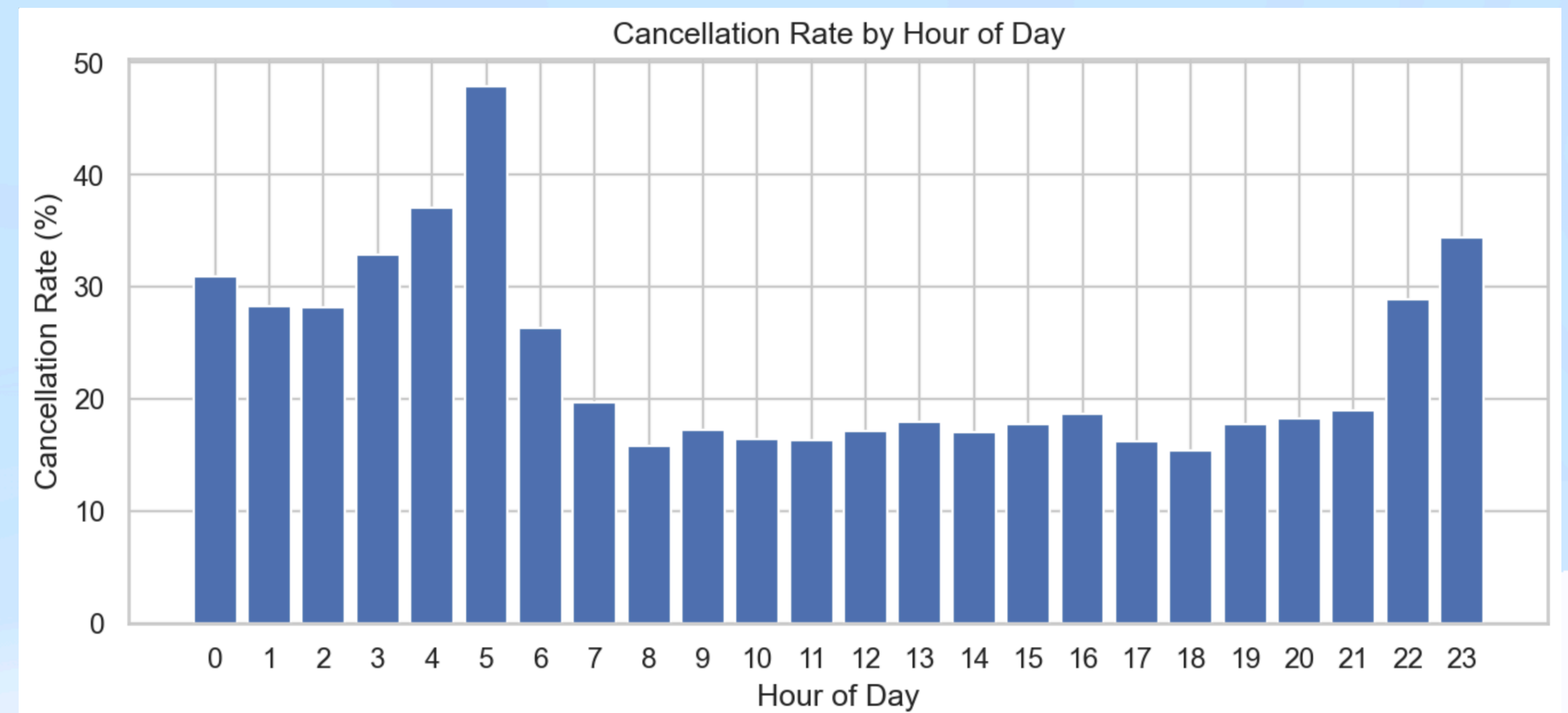
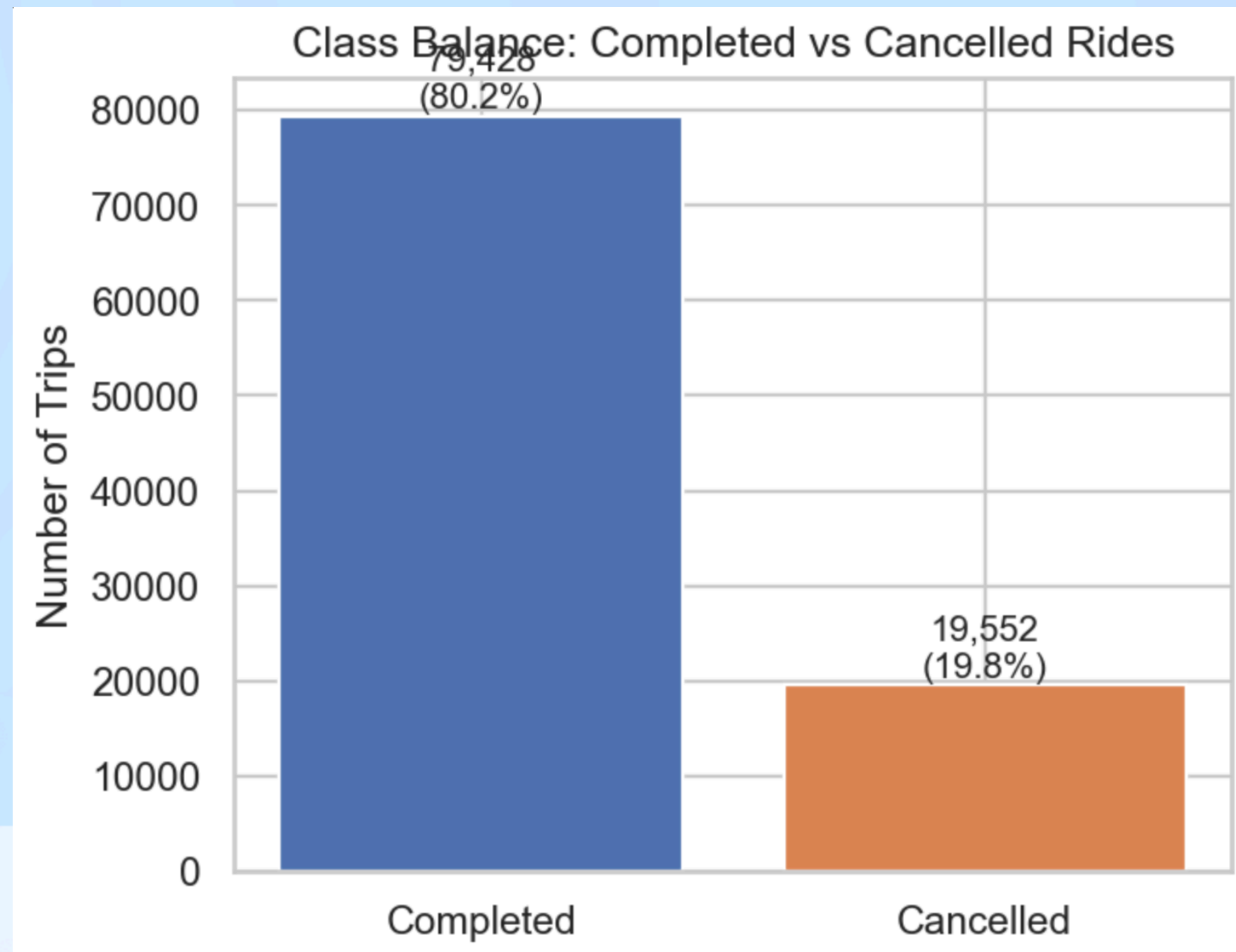
Dataset Overview

- Source: NYC TLC Yellow Taxi Trip Records January 2024
- 99,980 real trips - pickup time, zone ID, distance, fare amount
- Applying our labeling formula (label engineering)
- **80.2%** completed trips - 79,428 rides
- **19.8%** cancelled trips - 19,552 rides
- This 80/20 split matches realistic urban rideshare cancellation rates
- Important - a model that just always says not cancelled would be 80% accurate - but completely useless
- This is why we need precision, recall, and F1 - not just accuracy

How We Created the Label?

- We encoded 3 domain signals into a mathematical formula:
- Late night (10pm–5am) - drivers are scarce, cancellations spike - weight 1.0
- Long trip distance - higher cost and time, drivers reject more - weight 0.55
- Weekend - demand surges cause supply mismatches - weight 0.35
- Added Gaussian random noise so the model cannot just memorize the formula
- Passed through a sigmoid function to get a probability between 0 and 1
- Ran a 40-iteration search to calibrate exactly to 20% cancellation rate
- Each trip was then assigned cancelled=1 or cancelled=0 based on that probability

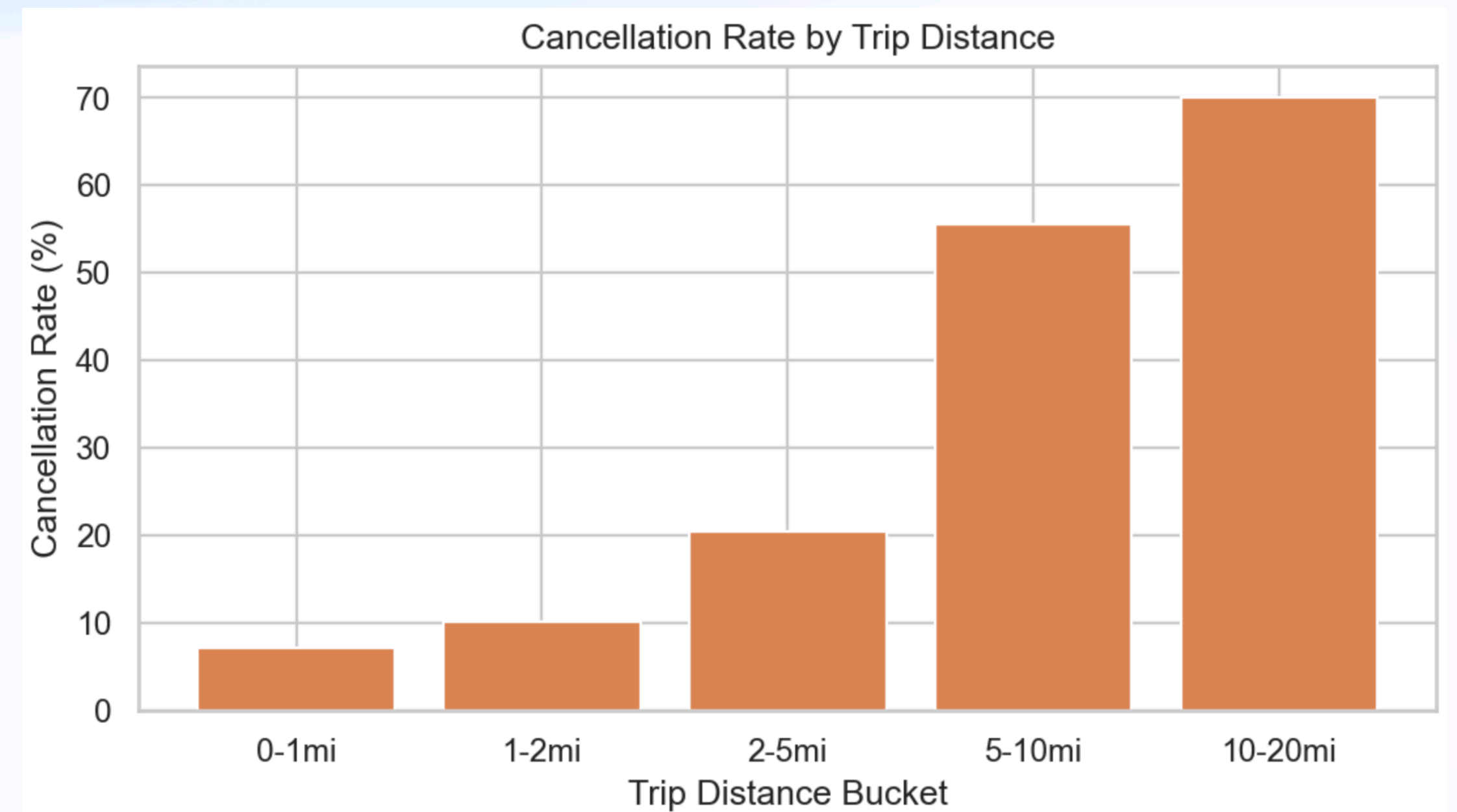
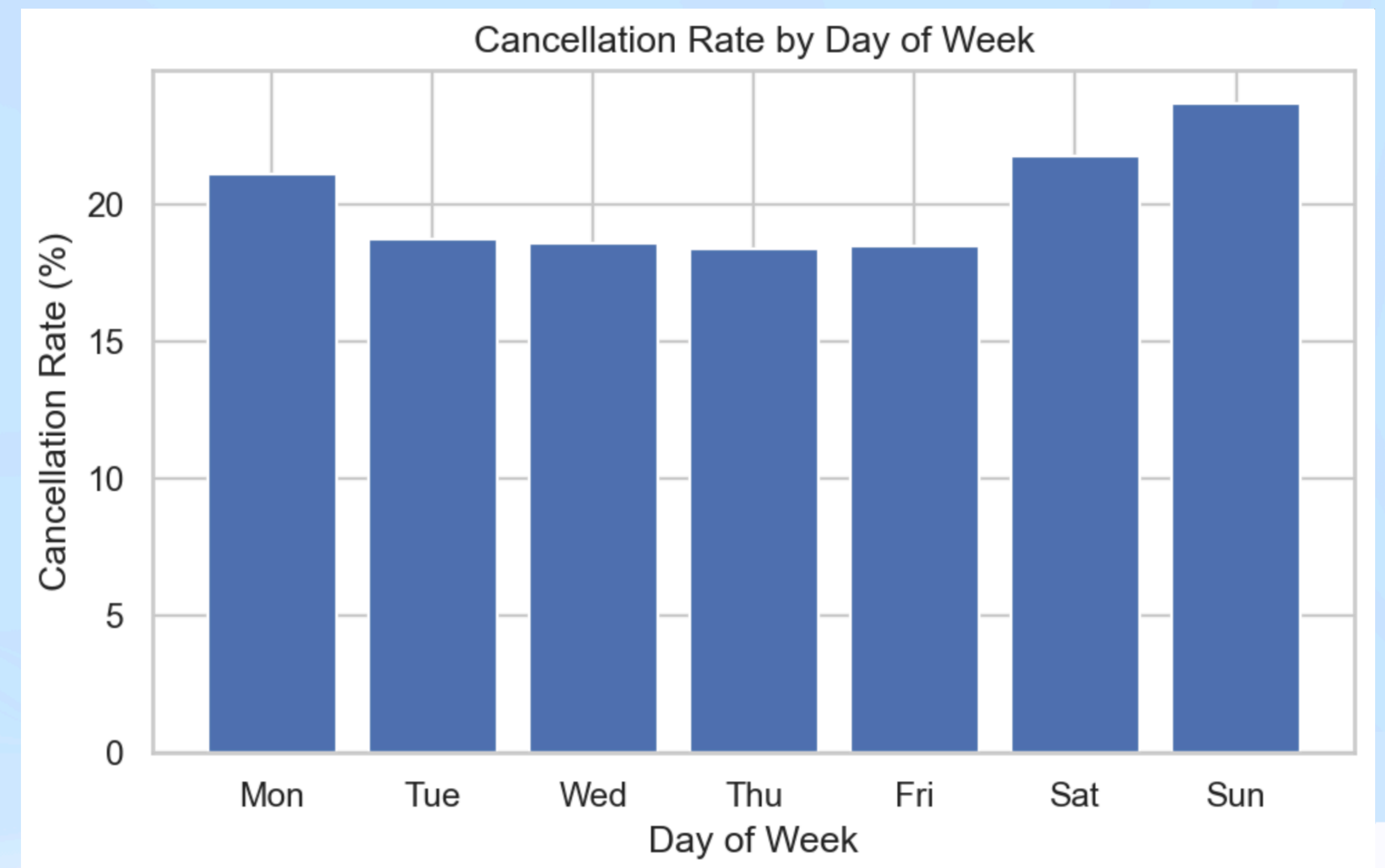
EDA - Class Balance + Cancel by Hour



- 80.2% completed vs 19.8% cancelled - intentionally calibrated
- Hours 3, 4, 5 AM peak at 33% to 48% cancellation rate
- Normal daytime 8am–5pm is flat at 15-16% low and stable
- Evening 10pm–11pm climbs back to 29-34%
- This U-shaped pattern validates our night signal in the labeling formula

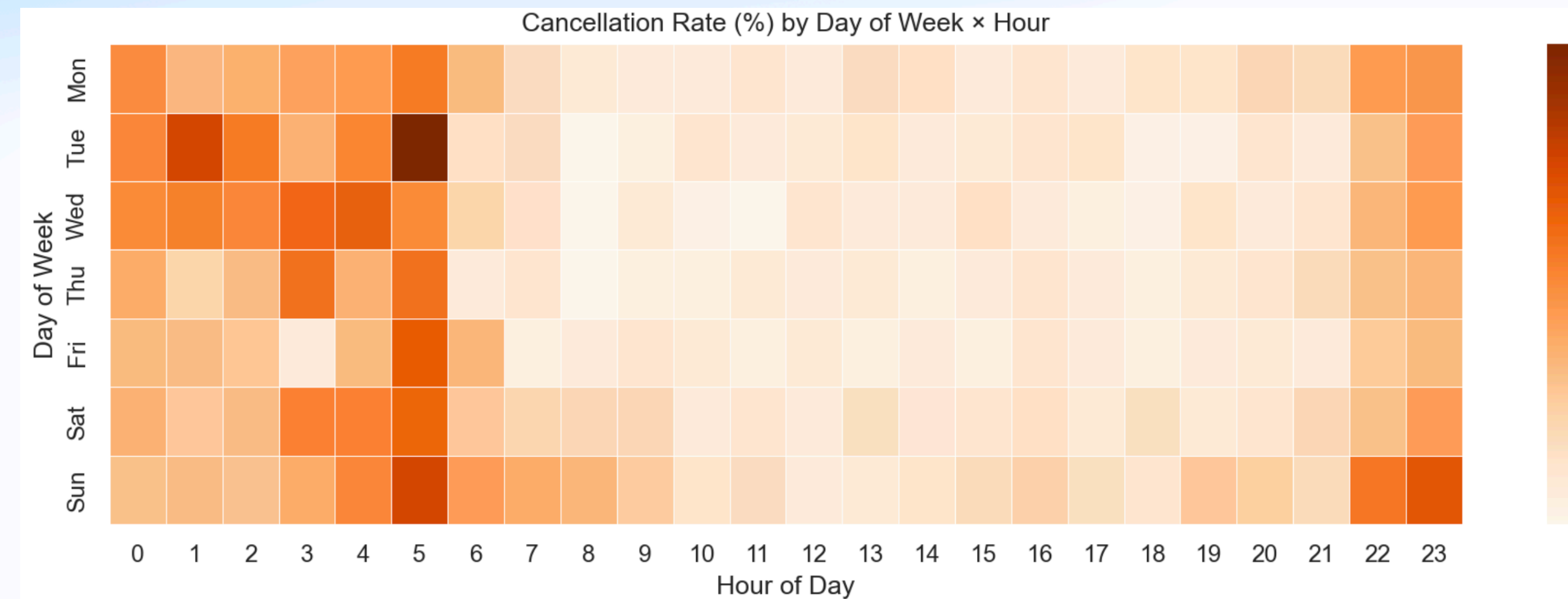
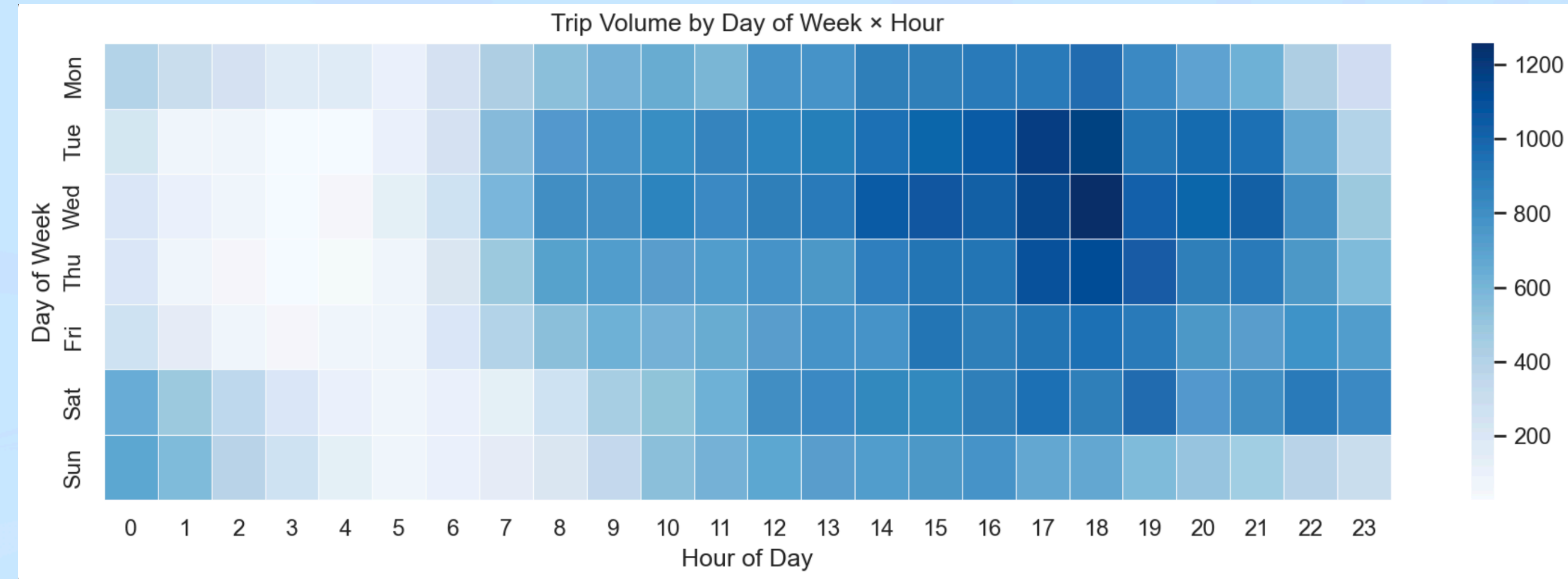
EDA: Cancel by Day + Cancel by Distance

- Sunday has the highest day rate at 24.6%, midweek sits around 18%
- Distance tells the most dramatic story in the dataset:
- 0–2 miles - only 4–6% cancel
- 5–10 miles - 67% cancel
- 10–20 miles - nearly 98% cancel
- One feature alone separates safe trips from risky ones almost perfectly
- This will become the most important finding in our ablation study



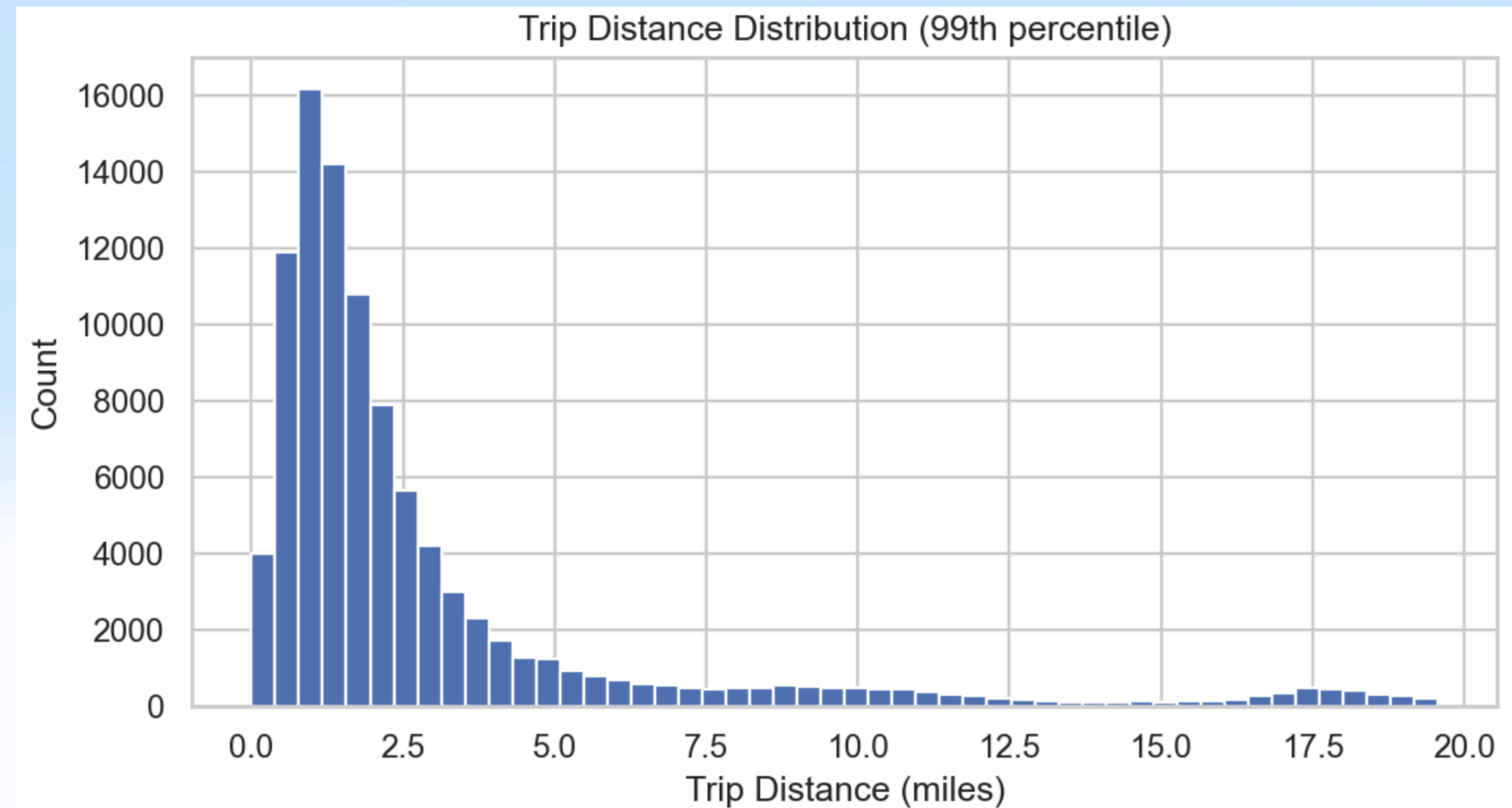
EDA: Heatmaps

- Sunday has the highest day rate at darkest blue = Wednesday 6pm–7pm - peak trip demand
- darkest red = 1am–5am on weekdays - peak cancellation danger
- The two heatmaps are nearly opposite - when most people ride, cancellation is lowest
- Sunday late night (22–23) : low volume and high cancellation rate
- Weekday early mornings (1–5am) are the highest risk window across the entire week
- hour and day cannot be collapsed into one feature - they interact differently

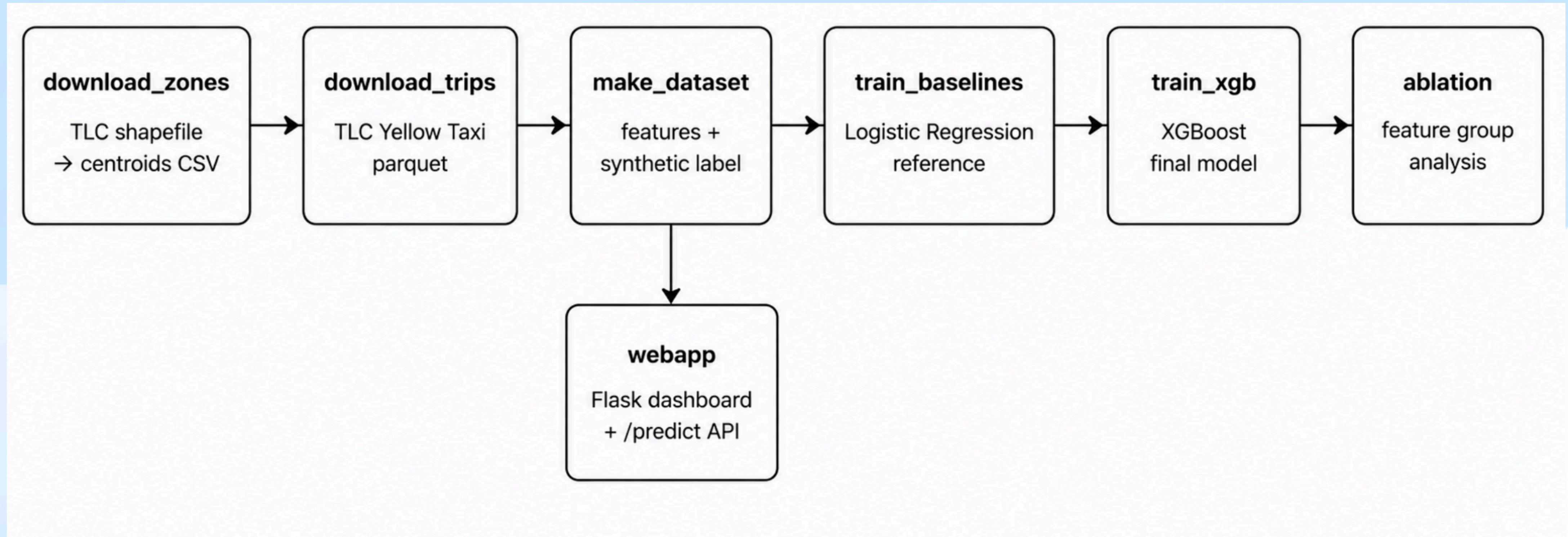


EDA: Trip Distance Distribution

- Majority of NYC trips are 1–2.5 miles short urban hops
- Distribution has a long right tail some trips reach 17–20 miles
- The model needs to handle both ends: short safe trips and long risky ones
- The tail (10+ miles) is where nearly all cancellations live
- This skewed distribution also means we needed to normalize distance before feeding it into the model



System Pipeline

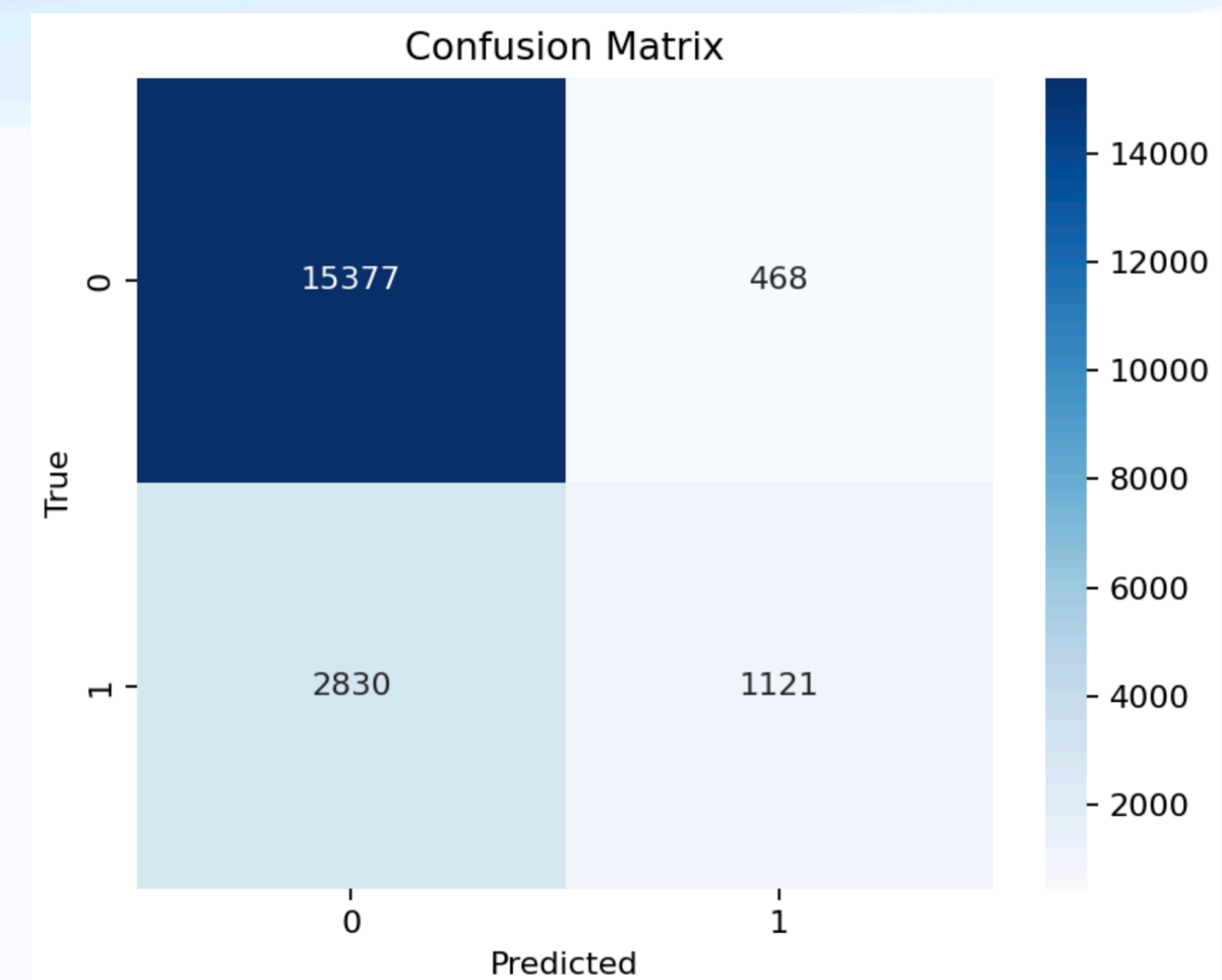
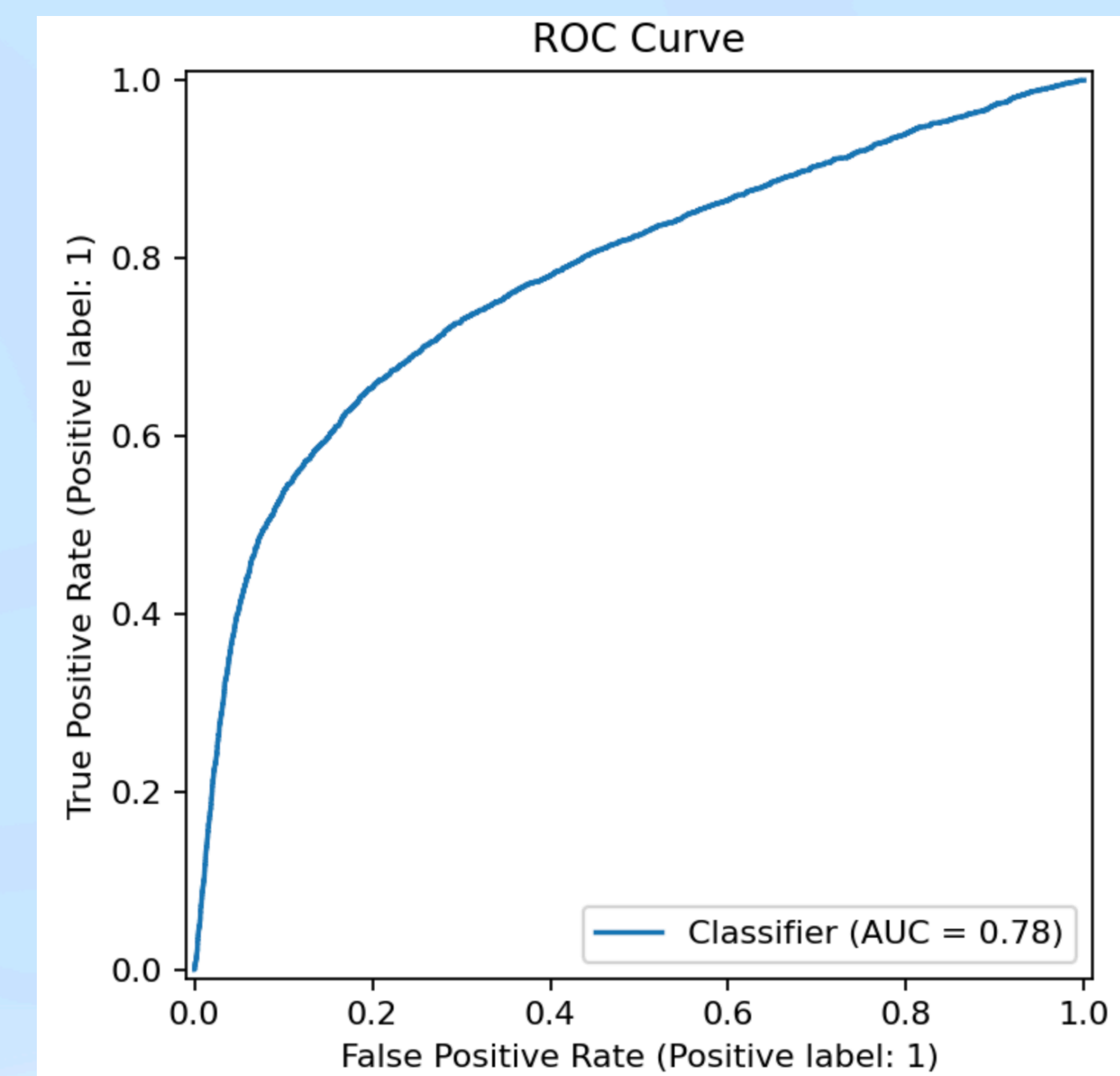


Feature Engineering

- 8 features total across 3 groups
- Temporal (2 features):
 - Pickup hour (0–23) — extracted from trip timestamp
 - Day of week (0=Monday to 6=Sunday)
- Spatial (4 features):
 - Pickup latitude & longitude
 - Dropoff latitude & longitude
 - Derived from official NYC TLC taxi zone shapefile → bounding box centroid → GPS conversion
- Trip (2 features):
 - Trip distance in miles — from TLC raw data
 - distance — straight line distance we computed between pickup and dropoff zones

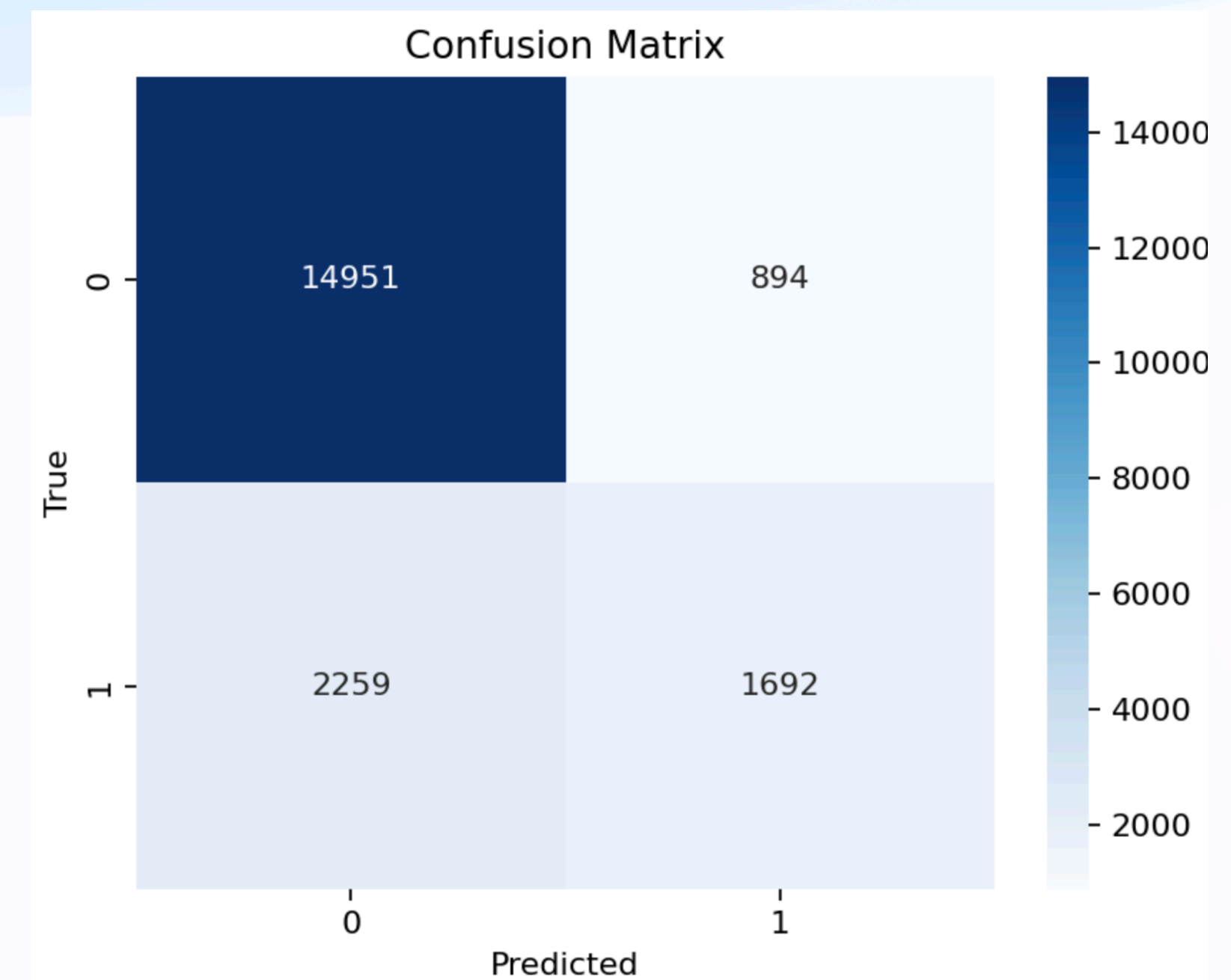
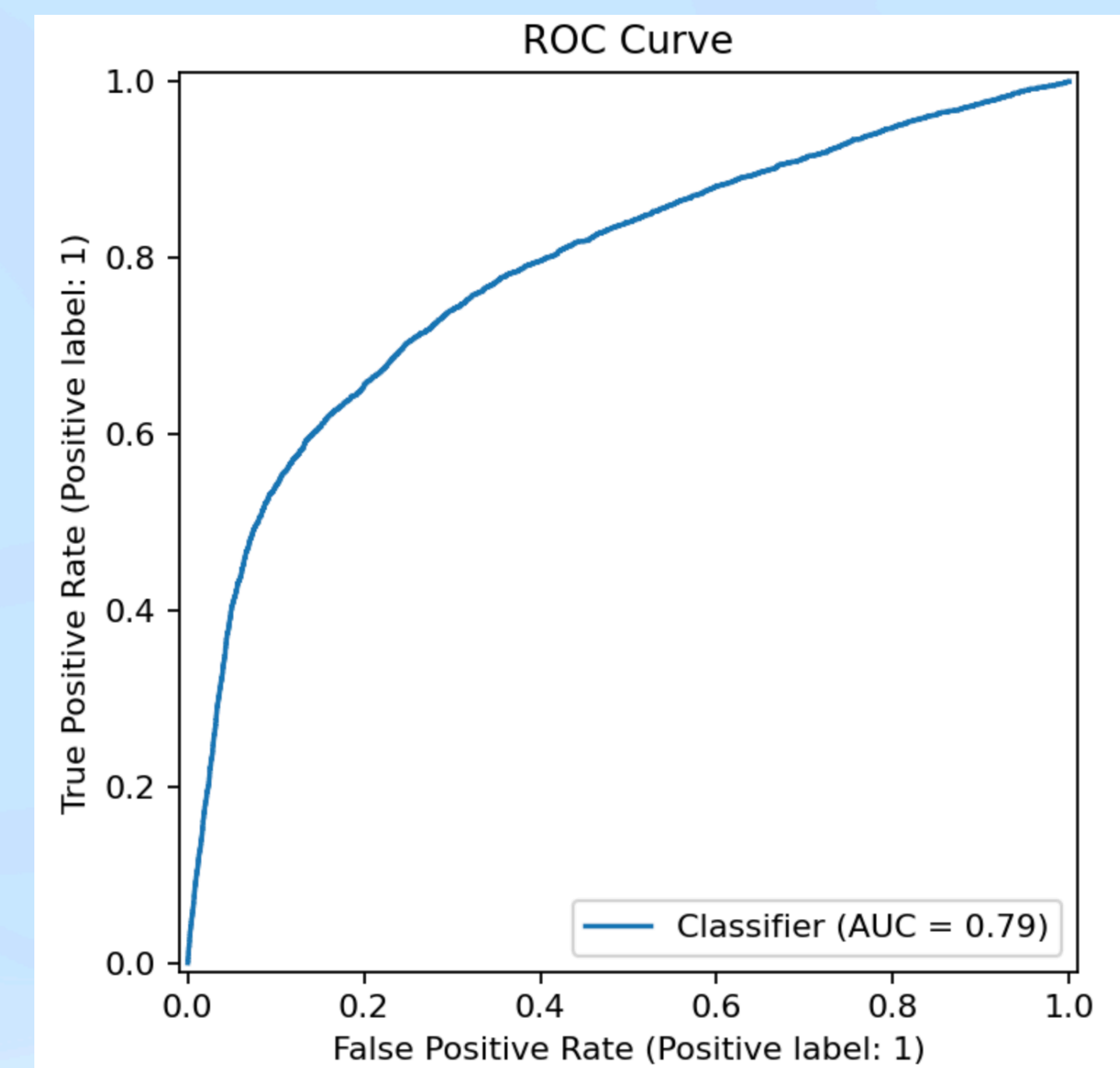
Model 1: Logistic Regression (Baseline)

- Linear model finds a straight boundary between cancelled and not cancelled
- class weight=balanced compensates for 80/20 imbalance automatically
- **ROC-AUC = 0.78** strong baseline
- Confusion matrix breakdown:
 - Caught **2,291** real cancellations of (58.6% recall)
 - **347 false alarms** out of 15,377 completed trips
- This is our reference point every other model is judged against this



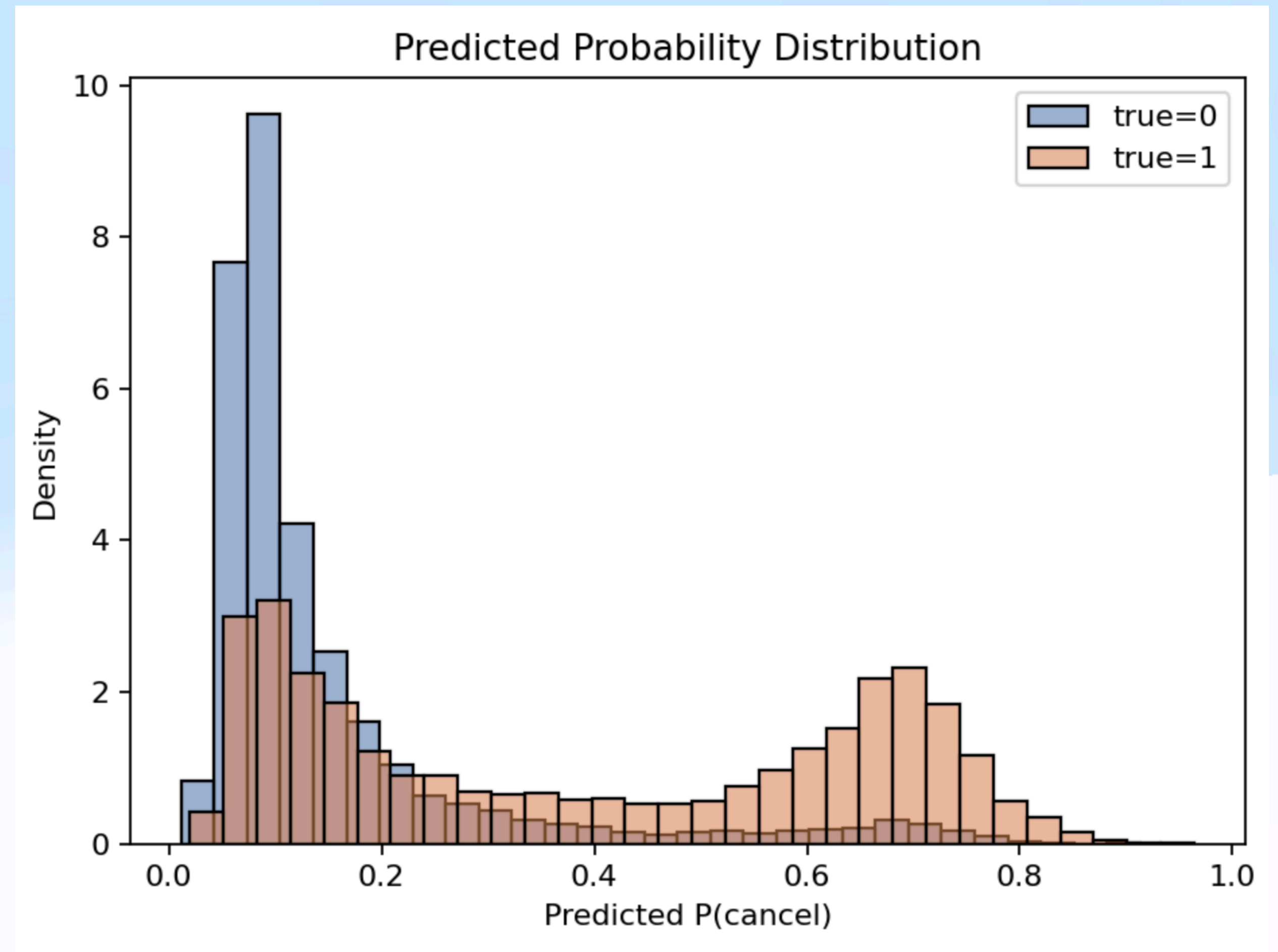
Model 2: XGBoost

- Builds hundreds of decision trees, each correcting errors of the previous
- Can capture non-linear patterns: e.g. long trip + late night + Sunday = very different from any single factor alone
- `scale_pos_weight = 4.0` penalizes missing cancellations 4 times more
- **ROC-AUC = 0.79** - better than baseline
- Confusion matrix breakdown:
 - Caught **2,259** real cancellations (60.1% recall) more than LogReg
 - **376 false alarms** - slightly more but catches 58 more real cancellations
- XGBoost wins on recall - the metric that matters most for dispatch



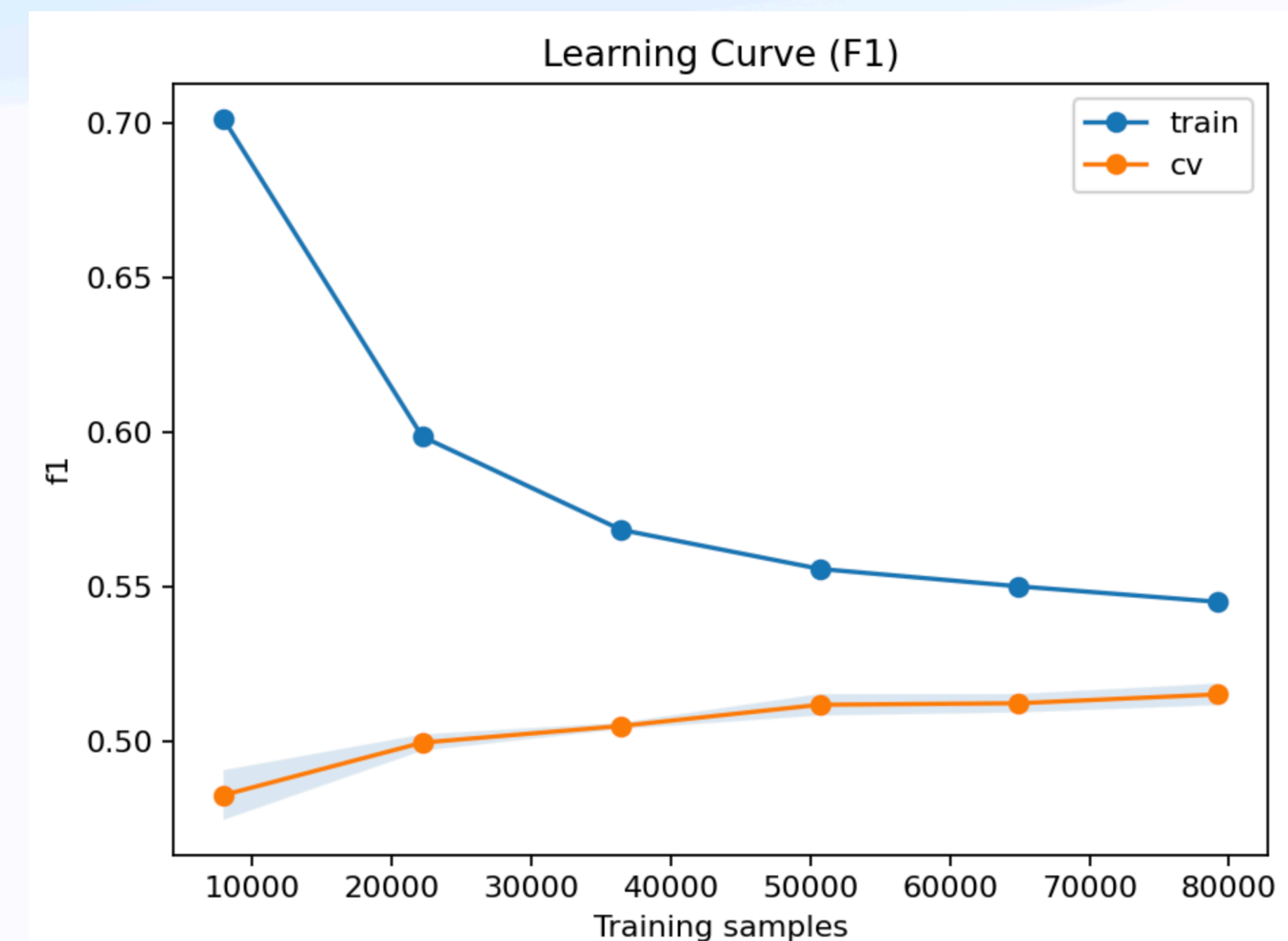
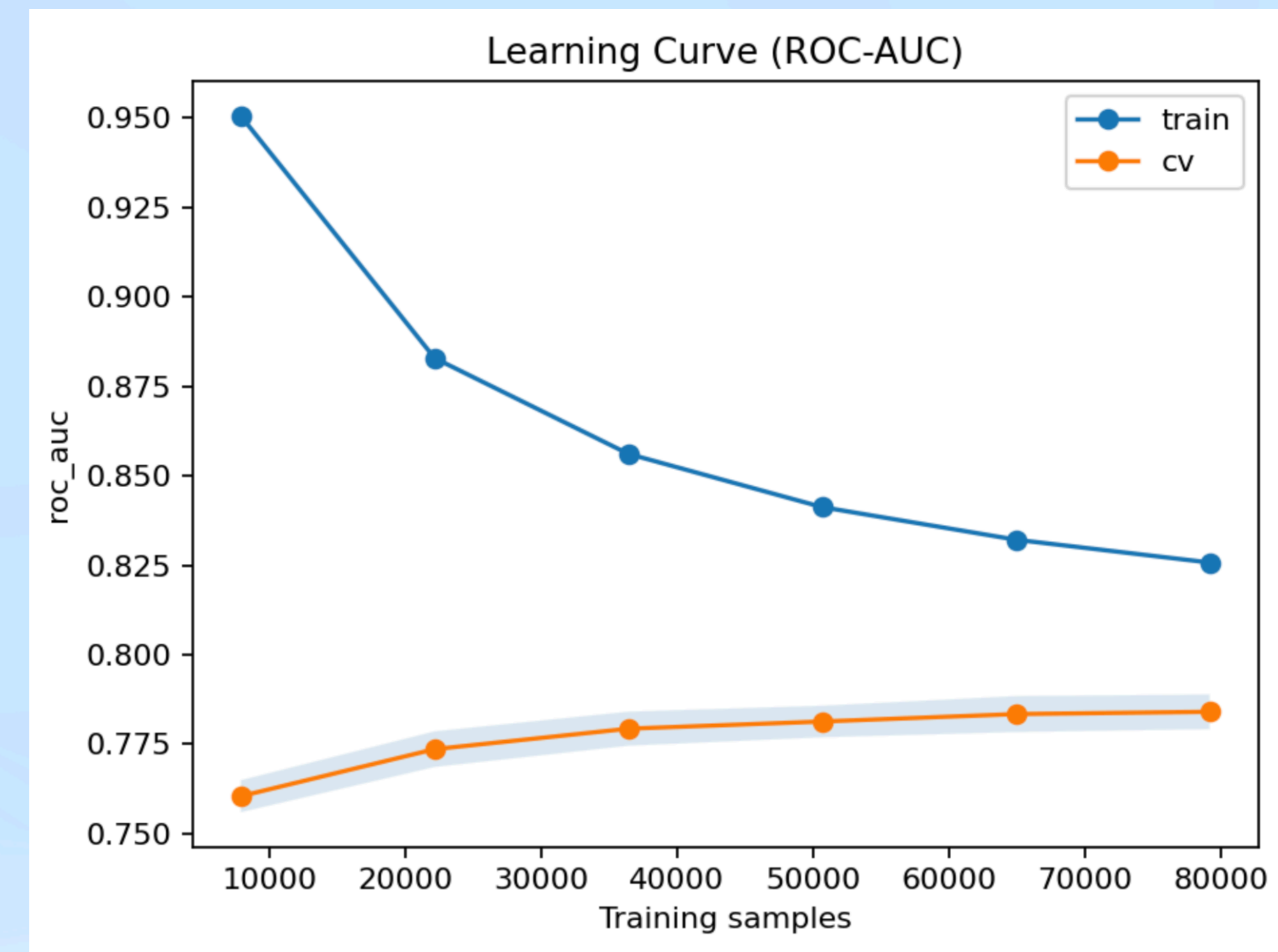
Prediction Distribution

- Builds hundreds of decision trees, each
cBlue bars = completed trips (true=0),
Orange bars = cancelled trips (true=1)
- Completed trips pile up near probability 0
model is very confident they will not cancel
- Cancelled trips pile up near probability 1
model is very confident they will cancel
- The clean separation at both ends is what
gives us $AUC = 0.79$
- The overlap in the middle (0.2–0.5) = the
hard uncertain cases the model struggles
with
- This chart proves the model has learned
real separable patterns, not random
guessing



Learning Curves

- Blue = training score, Orange = cross-validation score
- Training starts very high (model memorizes small data) then drops as data grows
- CV score starts lower but rises steadily and stabilizes near 0.78
- The two lines are converging at 80,000 samples - model is NOT overfitting
- F1 shows the same pattern - train and CV nearly meet at full data size
- Key insight: the CV curve is still slightly rising - more data would likely push AUC above 0.79



Ablation Study

- **Finding 1:** Temporal features alone = AUC 0.587 barely better than flipping a coin
- **Finding 2:** Trip distance alone recovers 87% of full model performance
- **Finding 3:** Removing spatial coordinates slightly improves AUC they add noise in small zones
- **Lesson:** Distance is the backbone. Time provides context. Location is secondary.

FEATURE SET	ROC-AUC	F1
all	0.7894	0.5169
temporal	0.5650	0.0000
spatial	0.7693	0.4949
trip	0.7837	0.5112
no_temporal	0.7802	0.5140
no_spatial	0.7909	0.5198
no_trip	0.7796	0.4996

Challenges We Faced

- **No cancellation label** — had to design the label from scratch, not just start modeling
- **Class imbalance** — 80/20 split means naive models always predict "not cancelled"
- **Zone coordinate conversion** — TLC shapefiles use NY State Plane projection, had to convert to GPS
- **Ablation surprised us** — we assumed time of day would matter a lot, but temporal-only AUC was 0.587
- **Distance dominance** — 68.5% feature importance was unexpectedly high, simplified our story significantly
- **Proposal said Neural Network** — we delivered XGBoost + HistGradientBoosting instead, which performed better with far less complexity

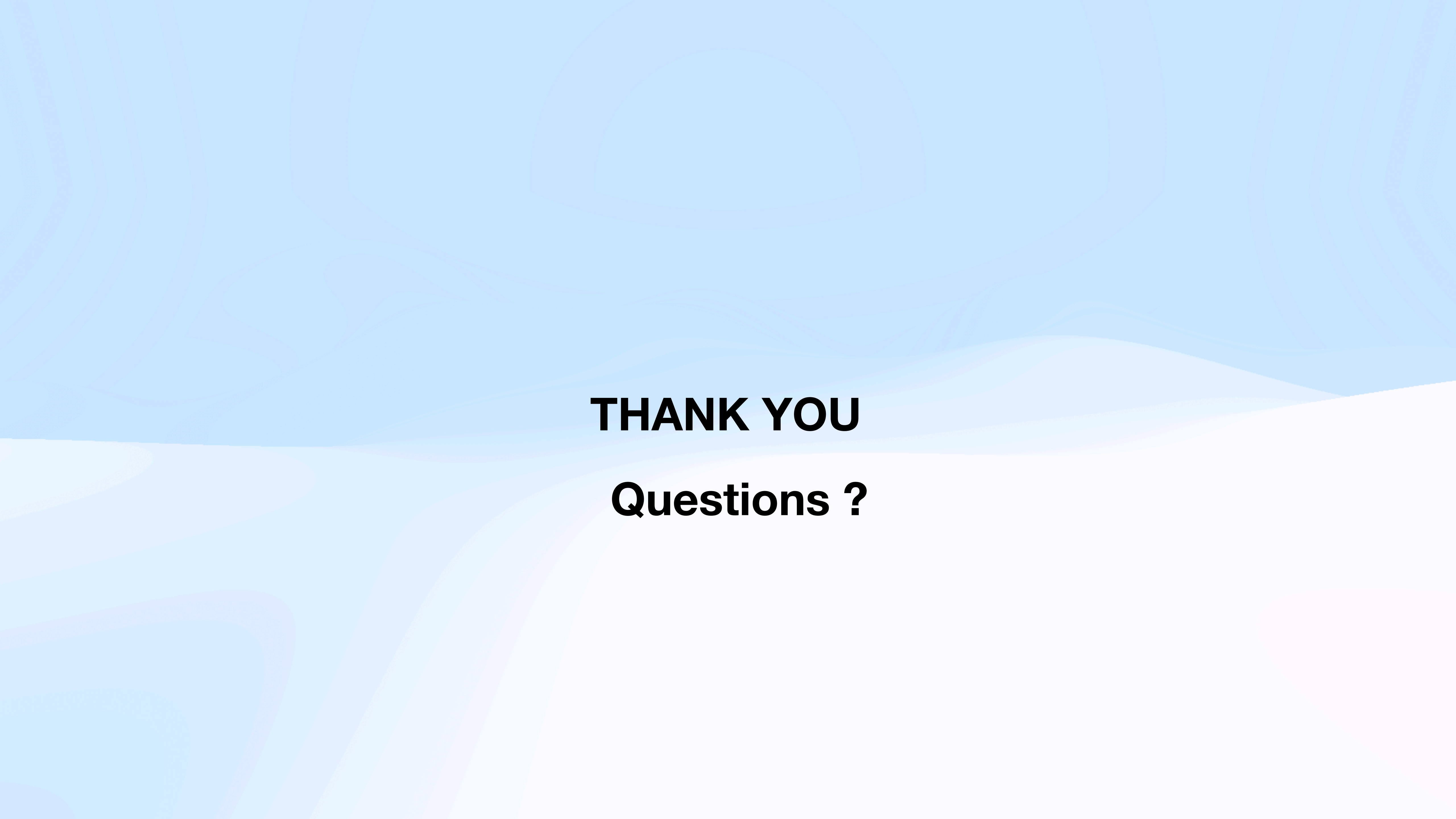
Conclusion & Future Work

Conclusion

- Ride cancellation can be predicted with $AUC = 0.79$ using only 8 features
- Trip distance is the dominant predictor accounts for 68.5% of model decisions
- Temporal features alone are nearly useless without distance context
- models agree the signal is real and consistent

Future Work:

- Get real cancellation data from a rideshare partner
- Add weather, rain and snow provably spike cancellations
- Add surge pricing, high fares cause rider-side cancellations
- rain across multiple months to capture seasonal effects



THANK YOU

Questions ?